

GDP: Measuring Total Production and Income

ECON 101 - Macroeconomics

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- 4.1 Gross Domestic Product Measures Total Production
- 4.2 Does GDP Measure What We Want It to Measure?
- 4.3 Real GDP versus Nominal GDP
- 4.4 Other Measures of Total Production and Total Income

Some Important Terms in Macroeconomics

- **Business cycle:** alternating periods of economic **expansion** and economic **recession**.
- **Expansion:** period during which total production and total employment are **increasing**.
- **Recession:** period during which total production and total employment are **decreasing**.
- **Economic growth:** the ability of an economy to produce **increasing quantities** of goods and services over the long run.
- **Inflation rate:** the **percentage increase** in the price level from one year to the next.

Why These Concepts Matter for GDP

- GDP is our main measure of the **size of the economy**.
- Over time:
 - Short-run **business cycles** show up as ups and downs in real GDP.
 - Long-run **economic growth** shows up as an upward trend in real GDP.
- Inflation affects **nominal** GDP, so we must distinguish it from **real** GDP.

4.1 Learning Objective

Goal

Explain how total production is measured.

- The most common measure of overall economic activity is **gross domestic product (GDP)**.
- **Definition:** Gross domestic product (GDP) is the **market value of all final goods and services produced in a country during a period of time**, typically one year.
- We will unpack each phrase in this definition:
 - “market value”
 - “final goods and services”
 - “produced in a country”
 - “during a period of time”

“Market Value”

- We produce many different goods and services:
 - cars, melons, haircuts, concerts, streaming services, etc.
- We cannot simply add *physical quantities* (e.g., 100 cars + 500 haircuts).
- We need a **common unit of measurement**.
- The most practical common unit is **money**.
- We value each good or service at its **market price**. This gives us the **market value** of production.

Final Goods and Services (1 of 2)

- GDP counts the market value of all **final** goods and services.
- A **final good or service** is purchased by a **final user**.
- These are the items included directly in GDP.
- **Intermediate goods and services** are inputs into other goods and services.
 - Example: a tire installed on a new truck, flour used in bread, steel used in cars.

Final Goods and Services (2 of 2): Avoiding Double Counting

- If we counted intermediate goods and also counted the final goods that use them, we would **double count**.
- Example:
 - A store buys ice cream from a manufacturer.
 - Later, the store sells that same ice cream to consumers.
 - If we counted both the store's purchase and the consumers' purchase in GDP, the value of the ice cream would be counted twice.
- To avoid this, we count **only the final sale** to the consumer.

GDP and Where Production Occurs

- GDP measures production within a **country's borders**.
 - It includes production by a Canadian firm or a foreign firm, as long as it takes place in Canada.
 - It excludes production that takes place outside Canada, even if it is done by Canadian-owned firms.
- Example:
 - A Japanese-owned auto plant in Ontario $\Rightarrow$ included in Canadian GDP.
 - A Canadian mining company producing in Chile $\Rightarrow$ not included in Canadian GDP.

“During a Period of Time”

- GDP is measured for a specific period:
 - usually a **year**, sometimes a **quarter** (three months).
- We count goods and services **produced during** that time.
- This avoids double counting across years.
- Example:
 - A textbook produced and sold new in 2022 is part of 2022 GDP.
 - The same textbook re-sold used in 2023 does **not** count again in 2023 GDP.
 - Used goods were already counted when they were first produced.

Production and Income: Two Sides of the Same Coin

- There are two main ways to measure total economic activity:
 - The value of **total production**.
 - The value of **total income**.
- When something is produced and sold, it generates **income** for someone:
 - workers receive wages,
 - owners of capital receive profit or interest,
 - landowners receive rent, etc.
- Therefore, in principle:

Total Production = Total Income.

Figure 4.1 – The Circular Flow and GDP (1 of 4)

- Start with a simple model of households and firms.
- Households:
 - Supply factors of production (labour, capital, natural resources, entrepreneurship).
 - Receive income in return (wages, interest, profit, rent).
- Firms:
 - Use these inputs to produce goods and services.
 - Sell output to households.

Figure 4.1 – Circular Flow: Basic Version

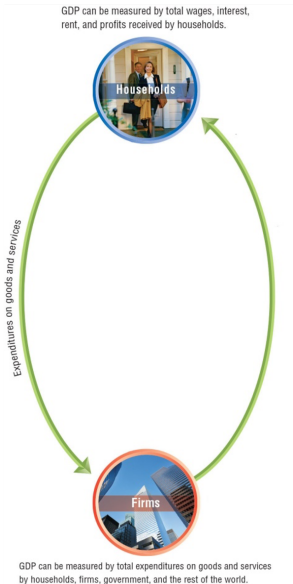


Figure 4.1 – Adding Government (2 of 4)

- Government collects **taxes** from households and firms.
- Government purchases goods and services from firms (G).
- Government makes **transfer payments** to households:
 - e.g., Employment Insurance, Old Age Security.
- Transfer payments are **not** payment for current production, so they are **not directly counted** in GDP.

Figure 4.1 – Adding Government

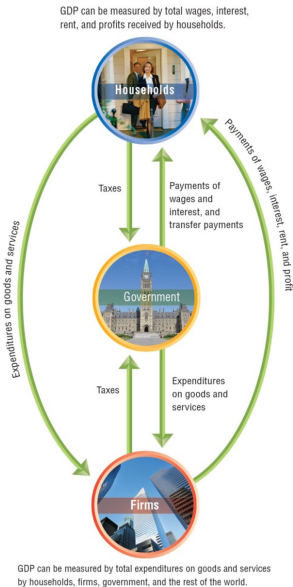


Figure 4.1 – Open Economy: Exports and Imports (3 of 4)

- Households, firms, and governments interact with the **rest of the world**.
- **Exports (X):**
 - Goods and services produced in Canada and sold abroad.
- **Imports (M):**
 - Goods and services produced abroad and purchased in Canada.
- Net exports: $NX = X - M$.

Figure 4.1 – Open Economy Flows

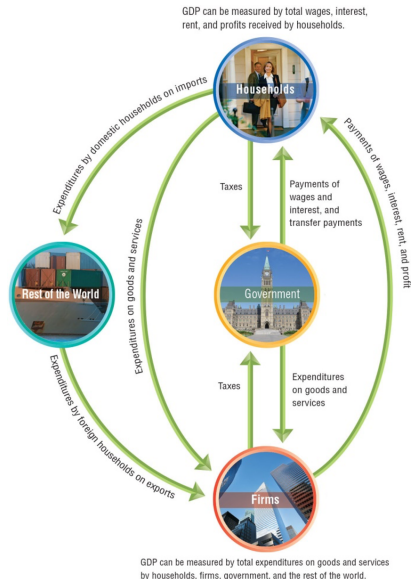
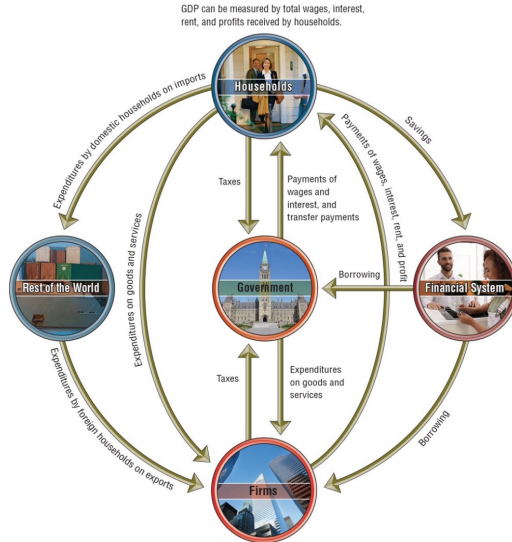


Figure 4.1 – Financial System (4 of 4)

- Households do not spend all of their income:
 - Some is saved with banks and other financial institutions.
- The **financial system** channels these savings to:
 - firms (for investment spending),
 - governments (for borrowing).
- This completes the circular flow of income and spending.

Figure 4.1 – Full Circular Flow



Income Approach to Measuring GDP

- One way Statistics Canada measures GDP is by tracking **incomes**.
- The income approach has **five components**:
 - Compensation of employees
 - Gross operating surplus
 - Gross mixed income
 - Taxes less subsidies
 - Statistical discrepancy

- **Compensation of employees** covers all benefits workers receive from providing their labour.
- It is the **single largest** element of GDP in the income approach, about **50%** of GDP.
- Two components:
 - **Wages and salaries:** money paid directly to workers ($\approx 44\%$ of GDP in 2021).
 - **Employers' social contributions:** payments made by employers on behalf of employees to social programs ($\approx 7\%$ of GDP in 2021).

- **Gross operating surplus** covers payments to owners of the **capital** used by firms and government.
- It accounted for about **27.7%** of GDP in 2021.
- Two elements:
 - **Net operating surplus:** payments to owners of capital over and above compensation for capital that wore out ($\approx 14.2\%$ of GDP in 2021).
 - **Capital consumption** (depreciation): payments to owners to cover capital that wore out ($\approx 13.4\%$ of GDP in 2021).

- **Gross mixed income** covers income generated by small businesses for their owners.
- It accounted for about **11.9%** of GDP.
- Two parts:
 - **Consumption of fixed capital:** depreciation of capital used by small businesses ($\approx 3.1\%$ of GDP).
 - **Net mixed income:** payment to owners over and above depreciation ($\approx 8.7\%$ of GDP).

- **Taxes less subsidies** are payments to government that act like income received by owners of other inputs.
- This component accounted for about **9.4%** of GDP.
- Intuition:
 - These are part of the value created in production, but they go to government instead of private input owners.

Income Approach Summary Table

Component	Approximate Share of GDP (2021)
Compensation of employees	50%
Wages and salaries	44%
Employers' social contributions	7%
Gross operating surplus	27.7%
Net operating surplus	14.2%
Capital consumption	13.4%
Gross mixed income	11.9%
Consumption of fixed capital	3.1%
Net mixed income	8.7%
Taxes less subsidies	9.4%
Statistical discrepancy	very small (around 0.0004%)

Expenditure Approach to Measuring GDP

- Another way to measure GDP is to track **spending** on final goods and services.
- The expenditure approach has **five components** in the national accounts:
 - Final consumption
 - Gross fixed capital formation
 - Investment in inventories
 - Net exports of goods and services
 - Statistical discrepancy

- **Final consumption:** all domestic purchases of goods and services by their final users.
- Includes purchases such as:
 - clothing,
 - paper by government,
 - food purchased by food banks.
- Final consumption is about **77%** of GDP.

Final Consumption by Sector

- Statistics Canada breaks final consumption into three elements:

Spender	Share of GDP
Households	54.1%
Government	21.4%
Not-for-profit organizations	1.5%
Total final consumption	77.0%

- **Gross fixed capital formation:** purchases of fixed assets (capital) by firms, governments, and households.
- Fixed assets:
 - provide benefits over a long period,
 - cannot easily be converted into cash.
 - Examples: buildings, machinery, equipment, software.
- Gross fixed capital formation is about **23.2%** of GDP.

Gross Fixed Capital Formation by Sector

Purchaser	Share of GDP
Business	19.1%
Government	4.0%
Total gross fixed capital formation	23.2%

- Business investment is crucial for **future productivity growth**.
- Government capital spending supports infrastructure, schools, hospitals, etc.

- **Investment in inventories:** spending on finished products kept on hand to sell later.
- If inventories rise, there is **positive** inventory investment.
- If inventories fall, inventory investment is **negative**.
- Example:
 - In 2021, inventory investment in Canada was about $-\$13,368$.
 - Firms were selling more out of inventory than they were adding.

Net Exports of Goods and Services

- **Net exports (NX)** = exports – imports.
- Exports:
 - Goods and services produced in Canada and sold to foreign buyers.
- Imports:
 - Goods and services produced abroad and bought by Canadian buyers.
- Net exports were about **2.8%** of GDP in 2021, and often **negative** for Canada:
 - Exports: 30.8% of GDP.
 - Imports: 30.6% of GDP.

Statistical Discrepancy (Expenditure Side)

- In practice, the income and expenditure approaches do not match perfectly.
- **Statistical discrepancy:**
 - An adjustment item that ensures the two approaches yield the same GDP.
 - Reflects small measurement errors and different estimation techniques.
- Statistics Canada:
 - Takes the difference between income-based and expenditure-based GDP,
 - Divides it in half,
 - Adds one half to the lower estimate and subtracts one half from the higher.

The Expenditure Approach in Macro Models

- In macro models, we often use a slightly different breakdown:

$$Y = C + I + G + NX$$

- **Consumption (C):**
 - Final consumption expenditure by households.
- **Investment (I):**
 - Business spending on fixed capital and inventories.
- **Government spending (G):**
 - Government fixed capital formation and final consumption.
- **Net exports (NX):**
 - Exports minus imports.

Components of GDP: Big Picture

- In Canada:
 - **Consumption** is the largest component of GDP.
 - Within consumption, **services** are the largest subcomponent (almost half of GDP).
 - **Investment** and **government spending** are smaller, but crucial for growth and public services.
 - **Net exports** are often slightly negative and reduce GDP.

Components of GDP

Figure 4.3 Components of GDP, Canada, 2021

Household consumption accounts for approximately 55.3 percent of GDP, far more than any of the other components. In recent years, net exports have typically been negative, which reduces GDP. Note that subtotals may not sum to the totals for each category due to rounding.

SOURCE: Statistics Canada. Table 36-10-0104-01 Gross domestic product, expenditure-based, Canada, quarterly ($\times 1,000,000$), <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610010401>.

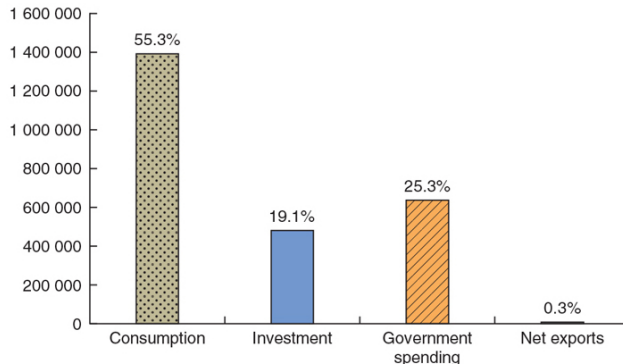


Table 4.1 – Calculating Value Added (Placeholder)

Table 4.1 Calculating Value Added

Firm	Value of Product	Value Added	
Diavik Diamond Mine	Value of raw diamond = \$500	Value added by diamond miner	= \$500
Gem cutter	Value of cut diamond = \$700	Value added by diamond cutter = $(\$700 - \$500)$	= \$200
Jewellery designer	Value of set diamond = \$1000	Value added by jewellery designer = $(\$1000 - \$700)$	= \$300
Jewellery store	Value of diamond ring = \$1500	Value added by jewellery store = $(\$1500 - \$1000)$	= \$500
	Total Value Added		= \$1500

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4.2 Learning Objective

Goal

Discuss whether GDP is a good measure of well-being.

- GDP is a useful measure of total output.
- Many people interpret GDP as a measure of citizens' **economic well-being**.
- However, GDP has shortcomings:
 - as a measure of **total production**,
 - as a measure of **well-being**.

Shortcomings of GDP as a Measure of Total Production

- Two important types of production are omitted:
 - **Household production**
 - **Informal (shadow) economy**

- Household production includes unpaid activities such as:
 - childcare,
 - cleaning,
 - cooking,
 - home repairs.
- If these tasks were performed by someone outside the household and paid for, they **would** be counted in GDP.
- Because they are unpaid, they are **omitted** from GDP, even though they create real value.

- The **informal economy** (or shadow economy) consists of:
 - Legal activities not reported to the government (to avoid taxes or regulations).
 - Illegal activities (e.g., some black-market transactions).
- These transactions create value and income but are not captured in official GDP statistics.
- Estimates suggest the Canadian shadow economy is about **13%** of officially measured GDP, and higher in many low-income countries.

How Serious Are These Omissions?

- For **year-to-year comparisons**:
 - Household production and the informal economy are likely roughly constant as a share of GDP.
 - So GDP growth is still a reasonable measure of growth in total production.
- Over **longer periods**, the omissions can matter more.
- Example:
 - As more women enter the paid labour force, tasks once done at home (childcare, cooking) are replaced by market activities.
 - GDP rises, but part of the increase reflects a **shift from home to market**, not entirely new production.

Shortcomings of GDP as a Measure of Well-Being

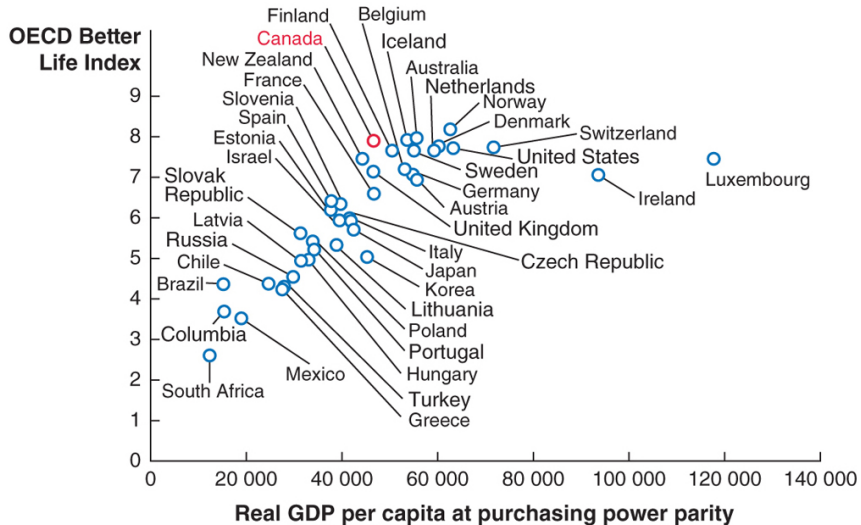
- GDP per person (**GDP per capita**) is often used as a measure of the **standard of living**.
- But even if GDP accurately measures total production, it does not capture:
 - the value of **leisure**,
 - **pollution** and other negative by-products of production,
 - **crime** and other social problems,
 - the **distribution of income**.

Example: Crime and Well-Being

- Suppose crime falls significantly.
- Less spending is needed on:
 - police,
 - prisons,
 - private security.
- GDP might **fall** because some spending drops.
- But economic well-being clearly **improves**.
- Conclusion: higher GDP is not the only indicator of better lives.

- GDP was not designed to measure happiness or life satisfaction.
- Yet, when we compare countries:
 - Higher GDP per person is often associated with higher measures of well-being (such as broader quality-of-life indices).
 - Very low GDP per person is almost always associated with low well-being.
- So:
 - GDP is an **imperfect** but **useful** indicator of economic well-being.

Better Life Index vs. GDP



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4.3 Learning Objective

Goal

Discuss the difference between real GDP and nominal GDP.

- GDP is measured in dollar terms (market value).
- When we compare GDP over time, prices usually **change**.
- We must separate:
 - changes in **quantities**,
 - changes in **prices**.

Nominal GDP vs. Real GDP

- **Nominal GDP:**
 - Value of final goods and services **evaluated at current-year prices.**
- **Real GDP:**
 - Value of final goods and services **evaluated at base-year prices.**
- The base year is arbitrary; in current Canadian data the standard base year is **2012.**
- Real GDP removes the effect of changing prices, allowing us to focus on changes in **quantities produced.**

Calculating Real GDP: Example (1 of 2)

- Suppose we have three goods, produced in 2012 and 2023.
- The table in the slides shows output and prices in 2012 and 2023.
- To calculate real GDP in 2023 using 2012 as the base year:
 - Multiply 2023 quantities by **2012 prices**, then add up.
 - The example yields:

$$\text{Real GDP}_{2023} = 4000 + 880 + 1800 = 6680.$$

Calculating Real GDP: Example (2 of 2)

- If we instead valued 2023 output at **2023 prices**, we would get:

$$\text{Nominal GDP}_{2023} = 7800.$$

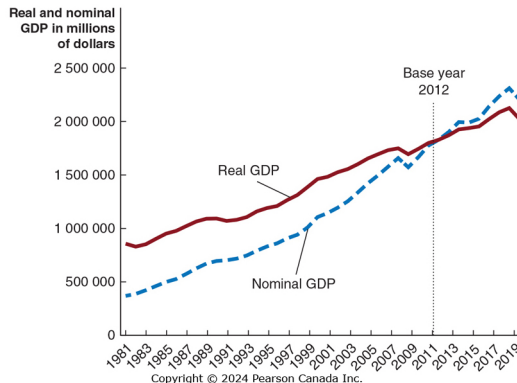
- Because most prices rose from 2012 to 2023, nominal GDP is larger than real GDP.
- This highlights why we use **real GDP** to avoid exaggerating growth when prices rise.

Figure 4.4 – Nominal GDP and Real GDP (Diagram Placeholder)

Figure 4.4 Nominal GDP and Real GDP

Currently, the base year for calculating real GDP is 2012. In the years before 2012, prices were, on average, lower than in 2012, so nominal GDP was lower than real GDP. In 2012, nominal GDP and real GDP were equal. Since 2012, prices have been, on average, higher than in 2012, so nominal GDP is higher than real GDP.

SOURCE: Statistics Canada, "Gross Domestic Product, Expenditure Based, Provincial and Territorial, Annual ($\times 1,000,000$)," Table 36-10.0222-01.



- In the base year, real and nominal GDP are **equal**.
- After the base year:
 - If prices in general rise, nominal GDP will grow **faster** than real GDP.
 - If prices fall, nominal GDP will grow **slower** than real GDP.
- Growth rates reported in the media usually refer to **growth in real GDP**.

Improving Real GDP: Chain-Weighted Prices

- One issue with a fixed base year:
 - **Relative prices** change over time.
 - Using old base-year prices can distort real GDP as we move further away from the base year.
- To reduce this distortion, Statistics Canada uses **chain-weighted** prices:
 - Uses previous-year prices to adjust current-year production.
 - This better reflects the changing structure of the economy.

The GDP Deflator

- Policymakers care about the **price level**: the average price of goods and services in the economy.
- Stable prices help households and firms plan for the future.
- One measure of the price level is the **GDP deflator**, defined as:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

- In the base year, nominal and real GDP are equal, so the GDP deflator is **100**.

Calculating the GDP Deflator: Example

- Suppose the GDP deflator increased from 108.8 to 109.0 between two years.
- The inflation rate based on the GDP deflator is:

$$\frac{109.0 - 108.8}{108.8} \times 100 \approx 0.2\%.$$

- We say the **price level** rose by about 0.2% between those years.

Other Measures of Total Production and Income

- GDP is the most common measure, but national accounts also construct:
 - Gross national income (GNI),
 - National income,
 - Personal income,
 - Disposable personal income.
- These measures adjust for:
 - income flows across borders,
 - depreciation of capital,
 - taxes and transfers to households.
- They are useful when we study **household budgets** and **long-run income dynamics**.

- **GDP** measures the size of an economy:
 - as total production **or** total income.
- GDP measures market activity and **omits** some important things:
 - household production,
 - informal economy.
- GDP is not intended to be a complete measure of well-being, but higher GDP per person is usually associated with higher material living standards.
- **Real GDP** controls for inflation; **nominal GDP** does not.
- The **GDP deflator** is a measure of the overall price level.
- National accounts also provide other measures of total production and income for more detailed analysis.

Thank you!